

Pools, Rules, and Tools: A Template-Integrated Resource Architecture

**A Meta-Project Demonstrating Fonds, Rules, and Tools
Integration in Research Software**

Field	Value
Author	Daniel Ari Friedman ¹
Affiliation	¹ Active Inference Institute
Correspondence	daniel@activeinference.institute
ORCID	0000-0001-6232-9096
Version	1.0.0
Date	2026-07-10
License	CC-BY-4.0
Repository	Public template repository
DOI	10.5281/zenodo.21298888
Keywords	research software engineering, monorepo architecture, reproducibility, fonds, governance rules, tool discovery, graceful degradation

Daniel Ari Friedman: Conceptualisation, architecture design, module implementation (fonds_reader, rules_applier, tools_invoker, integration, strong_rule_evaluator, figures, manuscript_variables), manuscript writing (all sections), test suite design, exemplar resource authoring (fonds, rules, tools), validation.

Acknowledgements I

The author thanks the Active Inference Institute for hosting the public template repository and providing the infrastructure within which this exemplar was developed. The design of the three-layer resource architecture draws inspiration from the Unix philosophy (Raymond 2003) and the enterprise application patterns documented by (Fowler 2002).

All source code, configuration files, and exemplar resources described in this paper are available in the public template repository under the `projects/templates/template_pools_rules_tools/` path. The integration pipeline is fully reproducible from source using `uv run python projects/templates/template_pools_rules_tools/scripts/02_run_integration.py` from the repository root. Generated manuscript variables are stored in `output/data/manuscript_variables.json` and injected at render time. Every figure in this manuscript, including the cover illustration, is likewise reproducible from source via `uv run python projects/templates/template_pools_rules_tools/scripts/05_generate_figures.py`, which writes directly to `manuscript/figures/`.

Competing Interests I

The author declares no competing interests.

Reproducibility Checklist I

This exemplar targets full computational reproducibility for every quantitative claim and figure it makes:

Reproducibility Checklist I

Artefact	Regenerated by	Verified by
Integration counts (fonds/rules/tools)	<code>scripts/02_run_integration.py</code>	<code>tests/test_integration.py</code> end-to-end assertion
Manuscript variable tokens	<code>scripts/03_generate_manuscript.py</code>	Rendered PDF contains no unresolved <code>{\dots}</code> tokens
All 9 figures (8 content + 1 cover)	<code>scripts/05_generate_figures.py</code>	<code>tests/test_figures.py</code> (per-function file-existence checks)
$\geq 90\%$ <code>src/</code> line coverage	<code>uv run pytest ... --cov-fail-under=90</code>	CI + local pre-push gate
Combined PDF	full project pipeline, PDF rendering stage (4-pass xelatex + bibtex)	<code>pdftotext</code> scan for unresolved-reference markers and page-scale raster read

Reproducibility Checklist I

A reader who clones the repository and runs the listed commands in order reproduces every number and image in this document without manual intervention. Quantitative claims in this document — fond/rule/tool counts, project inventory counts, and the runtime bar-chart figures (fig. ??, fig. ??) — are derived from the integration result, declared figure contracts, or the project filesystem at render time. None of these values are hand-typed; negative controls in `tests/test_manuscript_variables.py` change both integration inputs and a real temporary project tree to prove that token generation tracks its sources rather than emitting fixed constants.